

NOTICE

- 3 mins at most, I will reminder when 30s left.
- Tell us who you are by writing down your name on the board!
 - We need to know who we are voting for!
- Double check the doc file containing your is correct or not.

1 Bryce

2 Wisdom

3 Ben

4 Wesley

5 Adrian

6 William

7 Suman

8 Sharon

9 Cyrus

10 Emi

UI&UX, SERVICE DESIGN WITH AI

AI for Design

Yuan Liu - 2026 Spring

RECAP

- What is a brand
 - Brand design strategy
 - Brand visual design
 - Brand design with AI
-
- Technical part - How do our models process texts?

TODAY

- What is UI design
 - What is UX design
 - What is service design
-
- Technical part – what is a Transformer

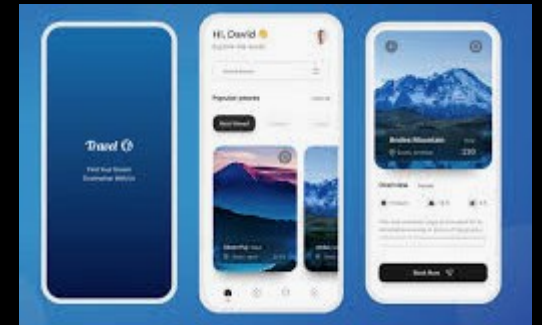
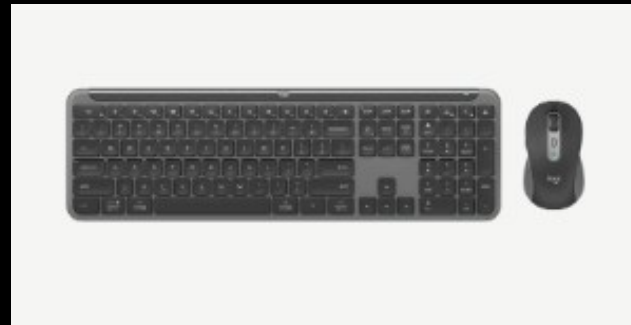
WHAT IS UI

AI for Design

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WHAT IS USER INTERFACE (UI)

- User interface (UI) refers to the space where interactions between humans and machines occur. It encompasses the design elements that allow users to interact with a device, application, or system.

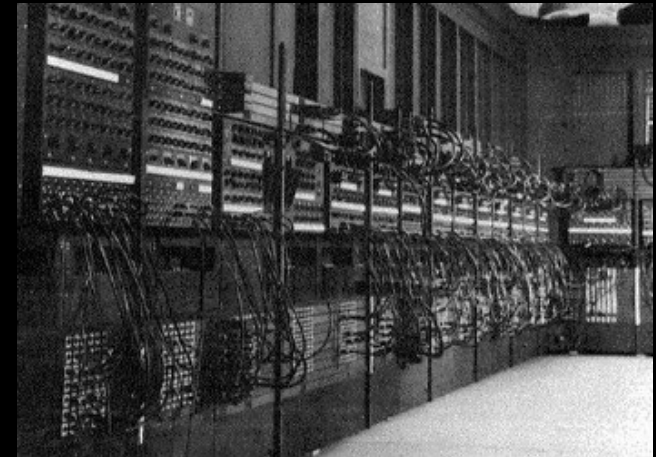


Discussion:

Could you provide some examples of user interfaces?

WHY UI EMERGED

- Complexity of Systems
- Human computer interaction (HCI)
- Accessibility
- User-Centered Design

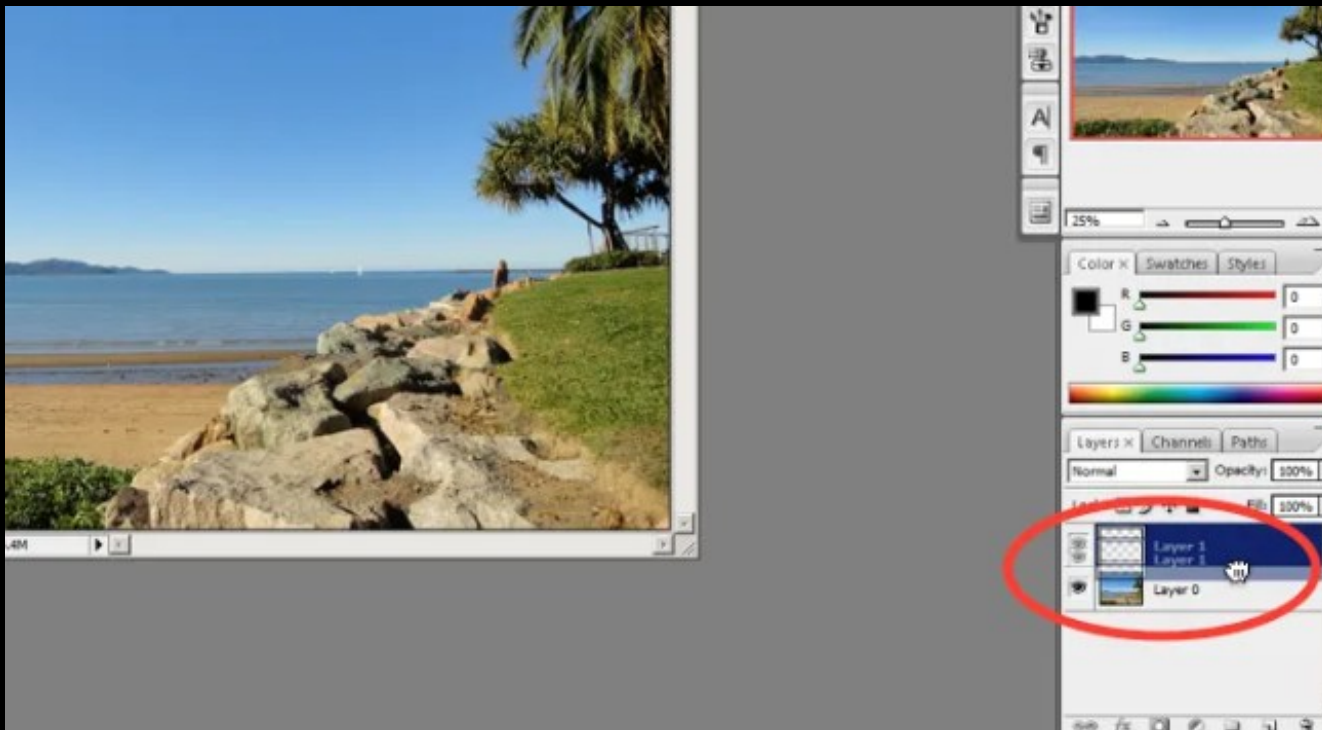


HOW TO DESIGN USER INTERFACES

- Users judge designs quickly and care about usability, utility and desirability.
- Usability refers to how easily and efficiently users can interact with a product or interface.
- Utility pertains to the functionality and features of a product—essentially, whether the product meets the needs of users.
- Desirability refers to the overall appeal and emotional connection a user feels towards a product.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

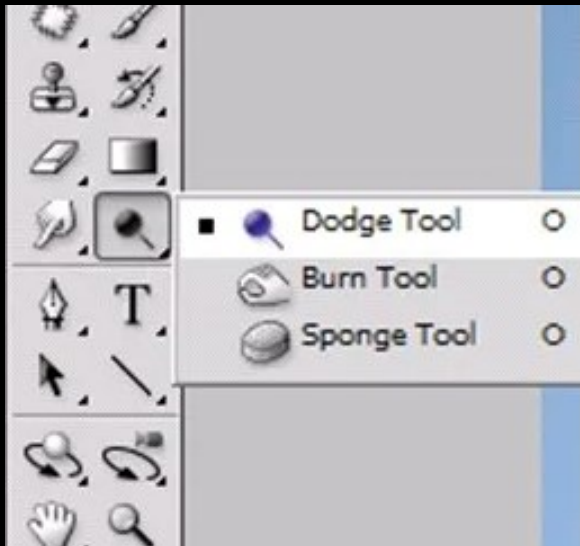
- **Visibility of system status.** Users should always be informed of system operations with easy to understand and highly visible status displayed on the screen within a reasonable amount of time.



When users move layers around in the Layers palette, they can visually see the layer being represented as physically dragged within the space.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

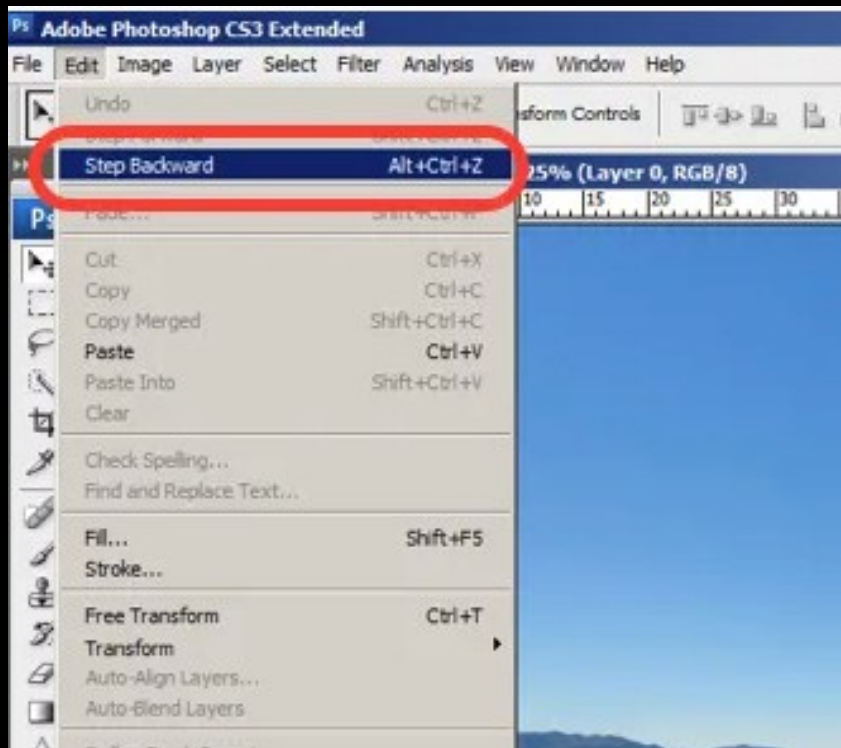
- **Match system and real world.** Mirror real-world language and concepts for target users. Present information logically, leveraging user expectations to reduce cognitive load and enhance usability.



Various tools like the dodge tool and the burn tool mimics a traditional darkroom technique for photographs.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

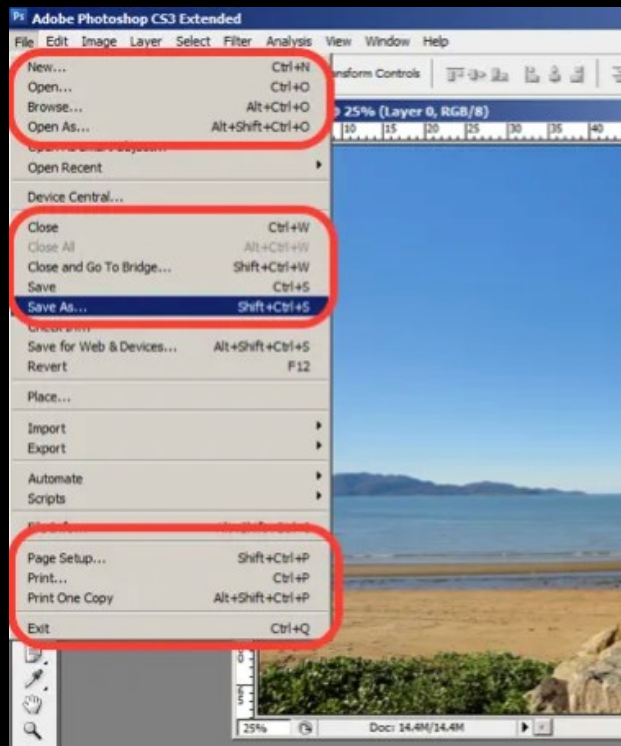
- User control and freedom. Offer users a digital space where backward steps are possible, including undoing and redoing previous actions.



As the user makes changes to an image or adds various artistic effects, they are able to quickly and easily take a step backwards if they make an error.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

- **Consistency and standards.** Maintain uniform graphic elements and terminology across platforms. Icons should represent the same concepts consistently throughout the interface.



PS maintains a standard layout and look and feel when it comes to the menu bar. They also utilize commonly known terminology such as “New...”, “Open...”, “Save As...”, etc.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

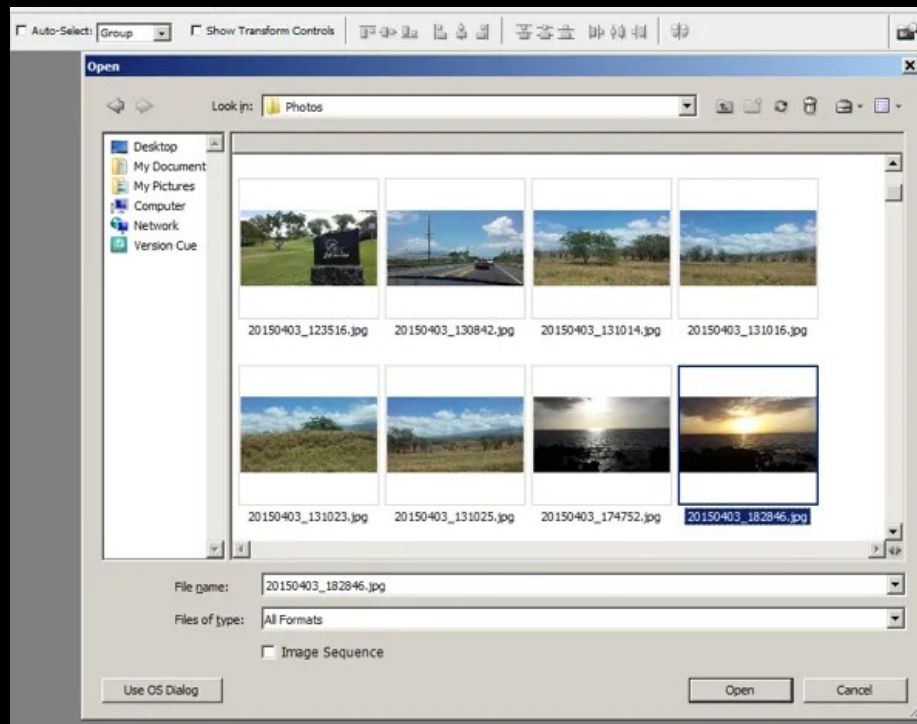
- **Error prevention.** Design systems to minimize potential errors. Eliminate or flag actions that may cause problems. Users prefer not to detect and fix issues, especially those beyond their expertise.



To prevent users from making errors, Photoshop provides a brief description or label of the tools when a user hovers over it to help make sure users are using the proper tool for the task at hand.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

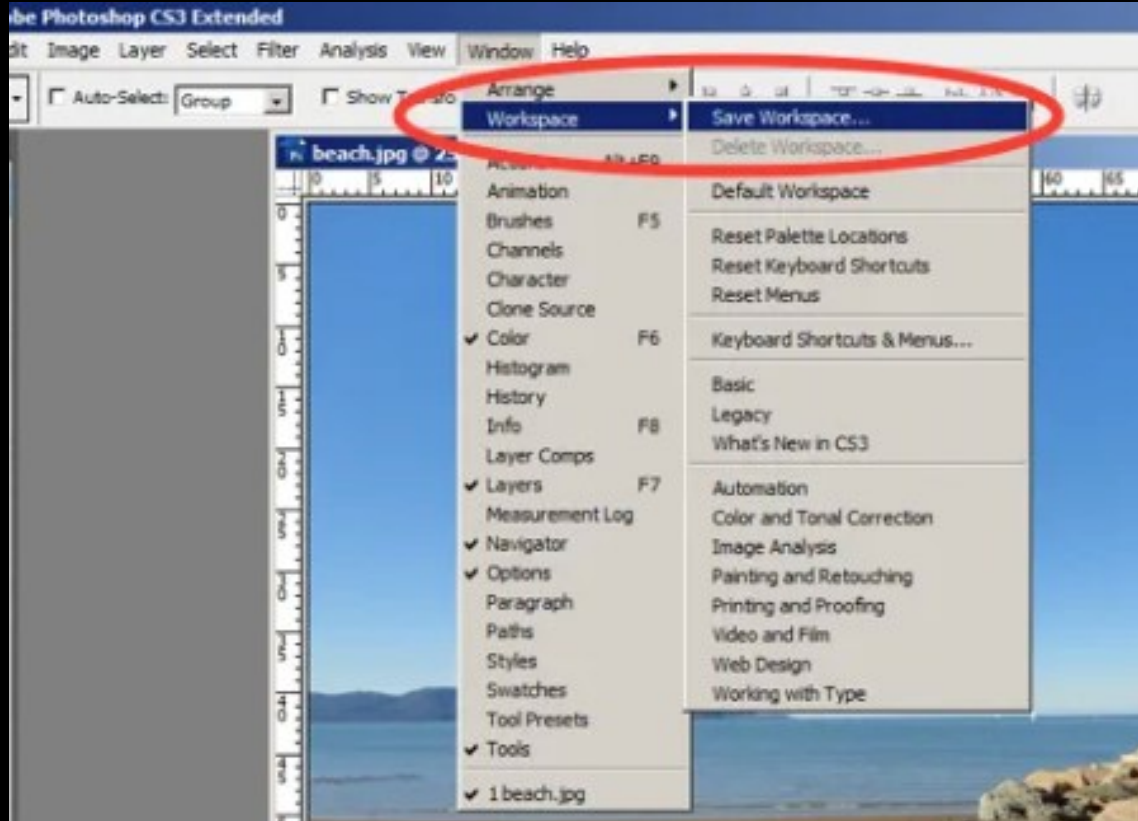
- **Recognition rather than recall.** Minimize cognitive load by keeping task-relevant information visible. Human attention is limited, with short-term memory holding about five items. Design for recognition over recall, as it's easier to perceive cues than remember information. For instance, multiple-choice questions are often easier than short-answer ones, as they require recognition rather than recall.



Whether it be making a selection from the artistic filters menu, or opening a new image file, Photoshop provides a sample view for users to make the right choice.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

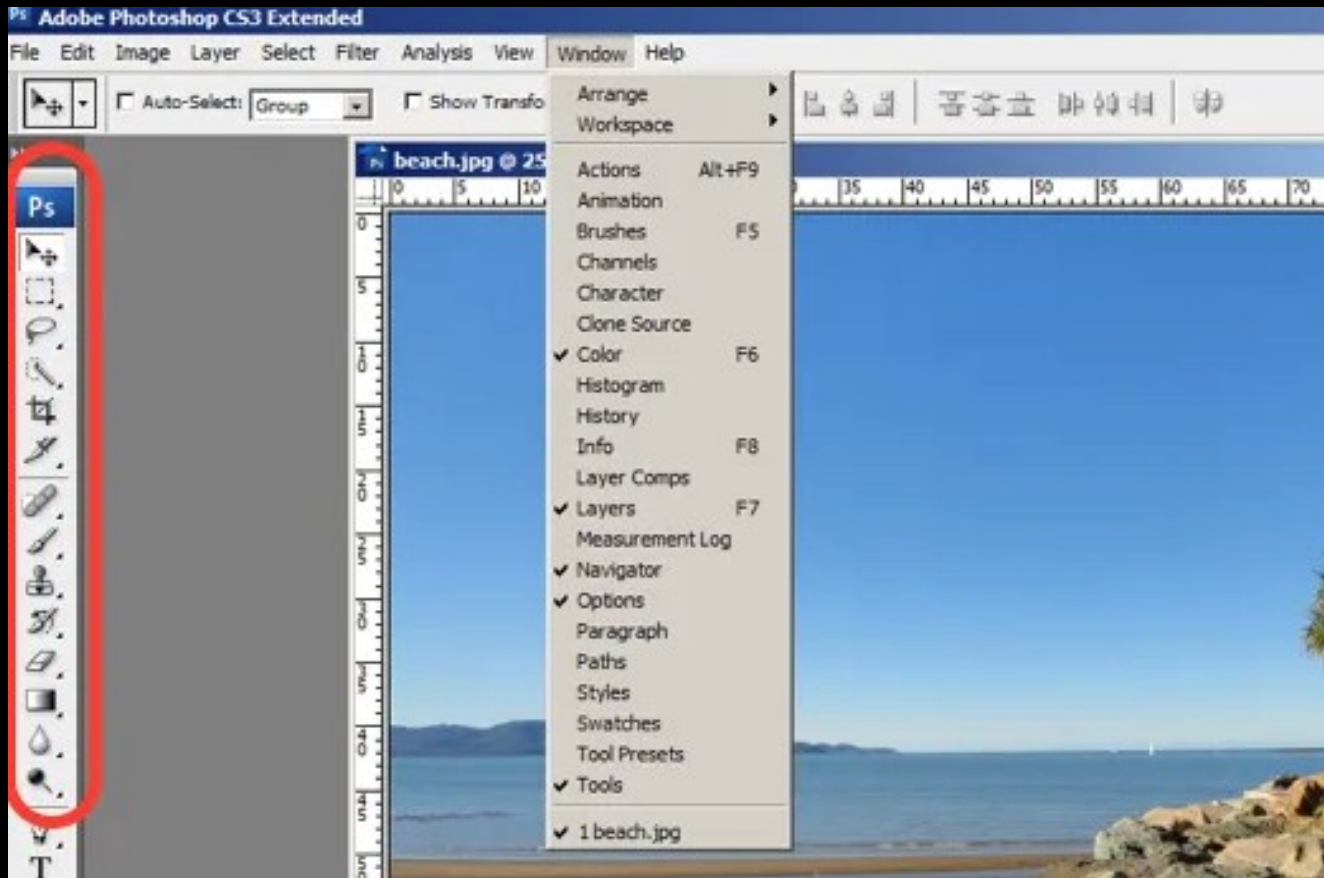
- **Flexibility and efficiency of use.** Enable faster navigation through customizable interfaces. Offer shortcuts like abbreviations and function keys for experienced users, while allowing newcomers to use standard interactions.



Users are able to utilize its flexibility by organizing and adding to their Workspace, as well as making things more efficient by saving it for future use.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

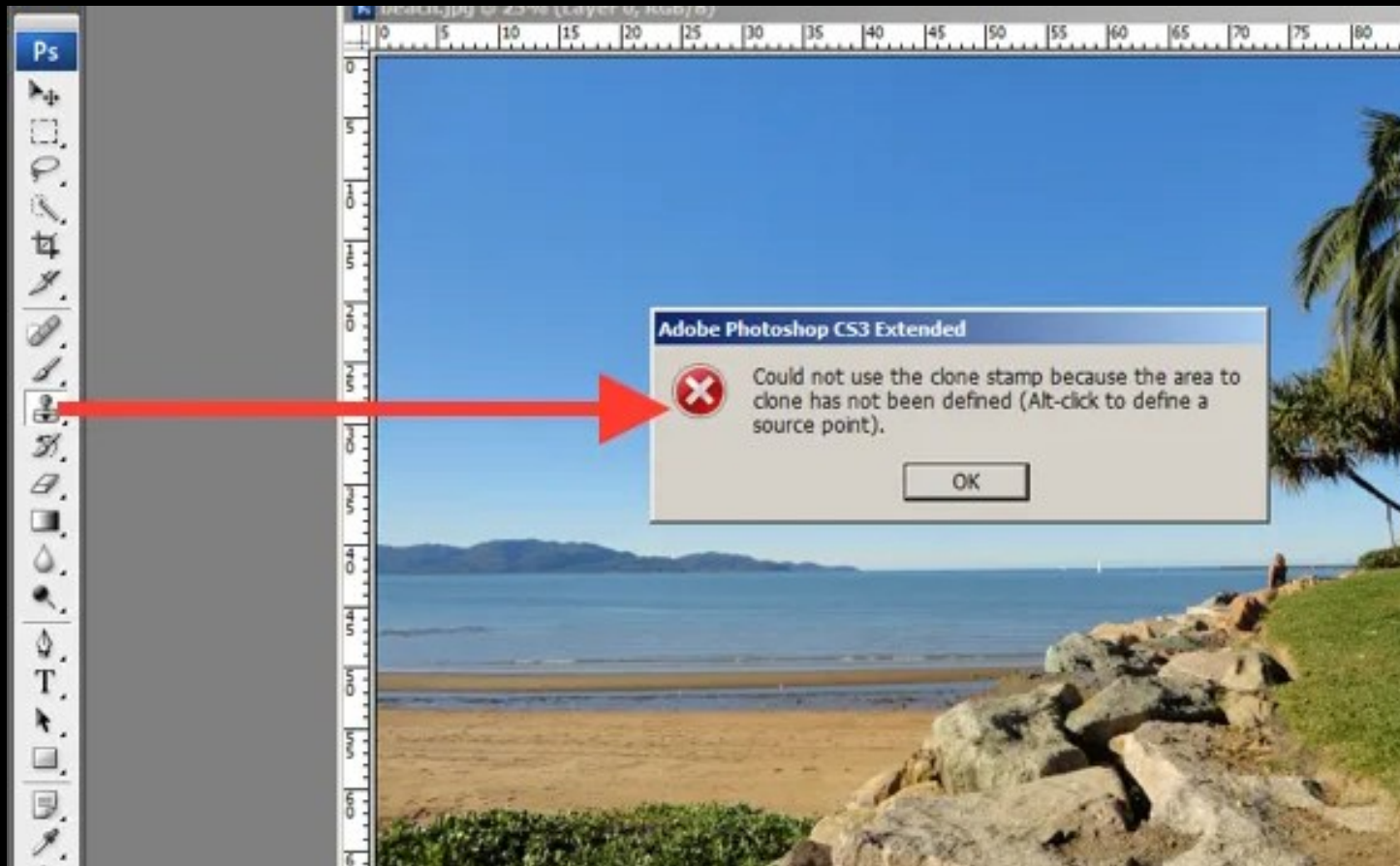
- Aesthetic and minimalist design. Minimize clutter to avoid overwhelming users. Display only essential elements for current tasks, while providing clear navigation options.



The toolbar in Photoshop only displays the icons and is neatly tucked to the side to help keep clutter to a minimum, and maintain a minimalist aesthetic.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

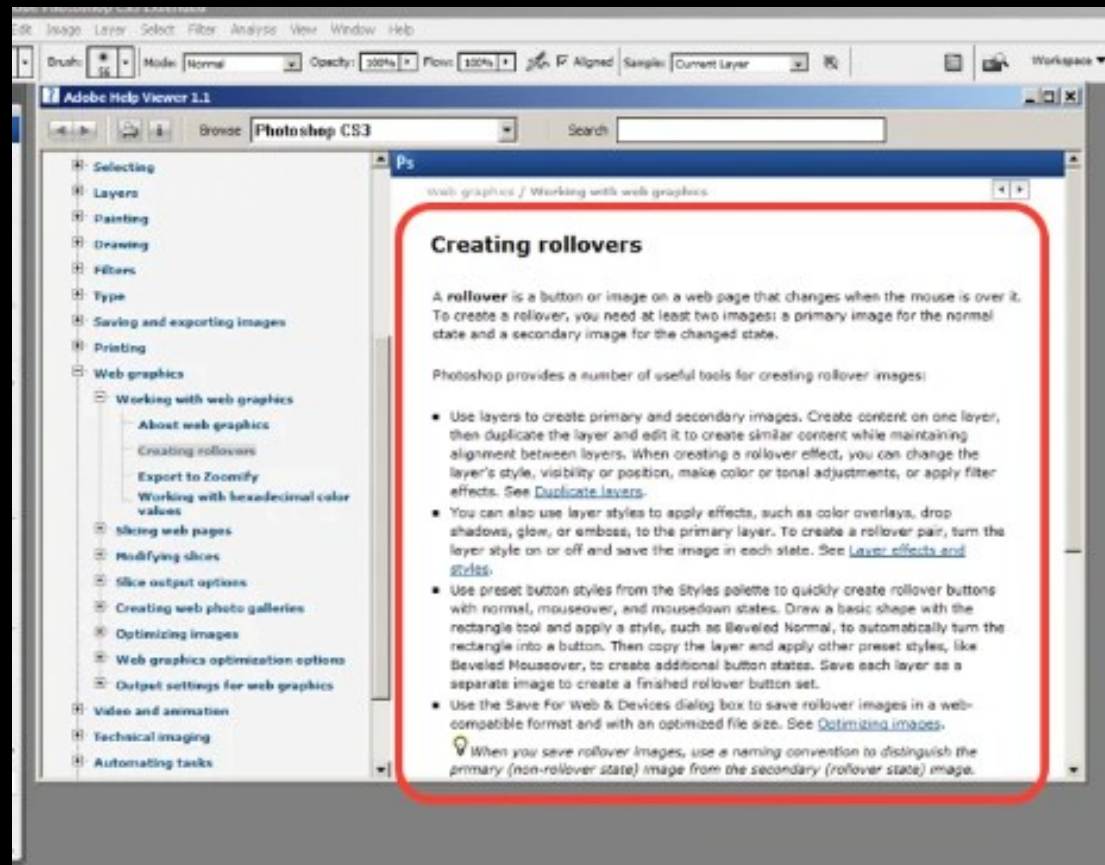
- Help users recognize, diagnose and recover from errors. Use plain language in error messages to ensure clear communication and easy understanding for all users.



Whenever there is an error, Photoshop provides dialogue that lets the user know what went wrong and how to fix it.

NIELSEN AND MOLICH'S 10 USER INTERFACE DESIGN GUIDELINES

- **Help and documentation.** While ideally users should navigate without documentation, it may be necessary. If required, ensure help is easily accessible, task-specific, and guides users to solutions effectively.



Help and documentation can be accessed easily via the main menu bar. From there, you can find a wide variety of help topics and tutorials on how to make full use of the program.

Analyse the SSC web UI (Part of Assignment)

[Home | Sustainable Smart Campus as a Living Lab \(hkust.edu.hk\)](http://hkust.edu.hk)

Rule of Thumb	Is this rule being applied? How so?	Is this rule violated? How so?	How can this rule further improve usability, utility and desirability?
1. Visibility of system status			
2. Match between system and the real world			
3. User control and freedom			
4. Consistency and standards			
5. Error prevention			
6. Recognition rather than recall			
7. Flexibility and efficiency of use			
8. Aesthetic and minimalist design			
9. Help users recognize, diagnose and recover from errors			
10. Help and documentation			

AT FIRST

- Define Goals

Identify Purpose: Determine the primary goals of the website

Target Audience: Understand who the users are and what they need from the site

- Put yourself in the user's shoes.

WHAT IS UX

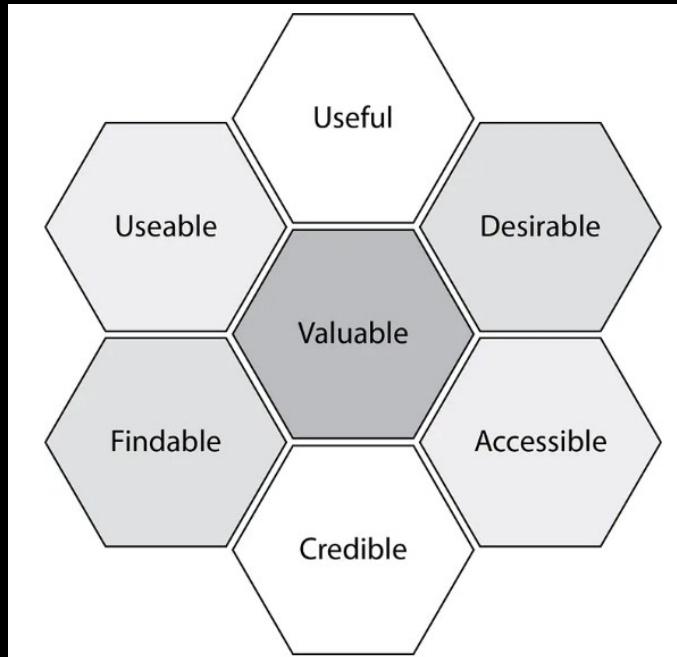
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WHAT IS USER EXPERIENCE(UX)

- User experience (UX) refers to the overall experience a person has when interacting with a product, system, or service.

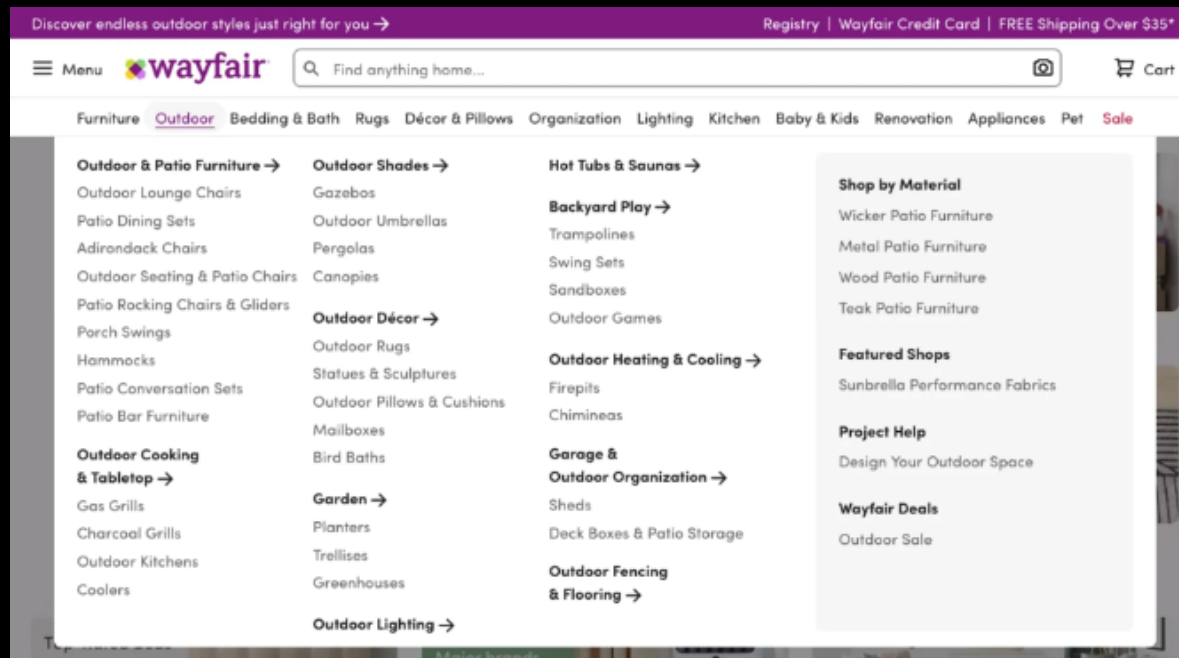
7 factors that describe user experience



Peter Morville's User Experience Honeycomb

7 FACTORS THAT DESCRIBE USER EXPERIENCE

- Usable: The system in which the product or service is delivered needs to be **simple and easy to use**. Systems should be designed in a way that is familiar and easy to understand. The learning curve a user must go through should be as short and painless as possible.



What problems of this page?

7 FACTORS THAT DESCRIBE USER EXPERIENCE

- Useful: A business's product or service needs to be useful and fill a need.
- Desirable: The visual aesthetics of the product, service, or system need to be attractive and easy to translate.
- Findable: Information needs to be findable and easy to navigate. If the user has a problem they should be able to quickly find a solution.
- Accessible: The product or services should be designed so that even users with a disabilities can have the same user experience as others.
- Credible: The company and its products or services need to be trustworthy.
- Valuable: The product must deliver value to the business which creates it and to the user who buys or uses it.

UX PROBLEMS

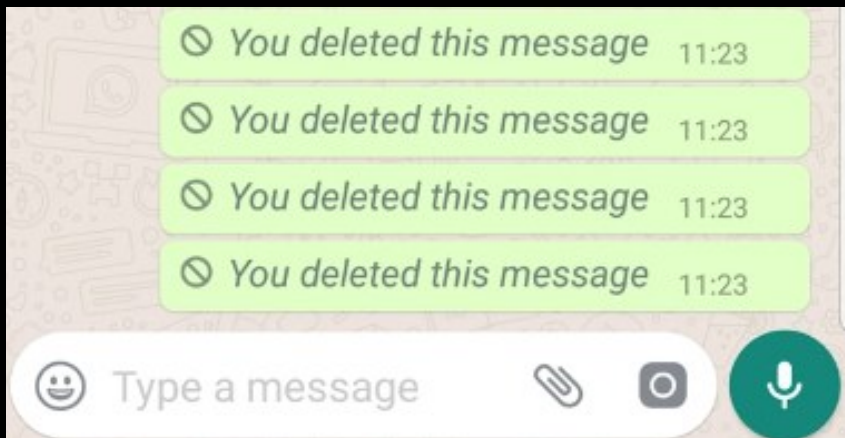
error- dialog.generic.header

error-dialog.generic.body

fatal-error.button-label

An issue arises when there's a critical failure, such as a login error. The app fails to provide clear, instructive feedback to the user in situations like these. A generic error message shows up informing the recipient, and that's it.

Informing the recipient that the sender has deleted a message somewhat defeats the purpose of deleting it in the first place. In fact, this looks way more suspicious and is likely to prompt an awkward “why did you delete the message?” type of response.

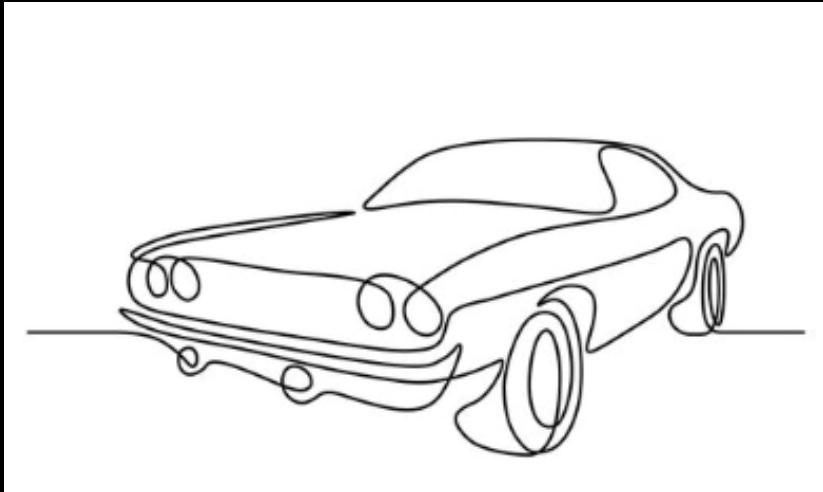


UX PROBLEMS

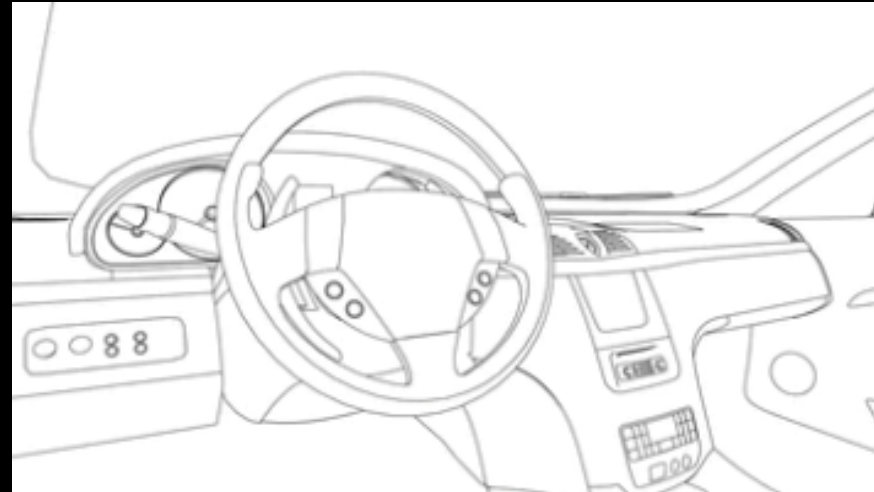


UI VS. UX DESIGN

- UX encompasses the **entire experience** a user has with a product, while UI is specifically concerned with the design of the interface through which users interact with that product.



UX designer:
How can we enhance the overall experience of driving, including comfort, safety, and convenience?



UI designer:
How should we arrange the controls and displays on the dashboard for easy access and visibility?

WHAT IS SERVICE

AI for Design

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WHAT IS SERVICE

- A service is an **intangible act or performance** delivered to meet the needs of users or customers. Unlike physical products, services involve a process and interaction between the provider and the consumer.
- Key characteristics of services include:
 - Intangibility: Services cannot be touched or owned; they are experienced.
 - Inseparability: Services are often produced and consumed simultaneously (e.g., a haircut).
 - Variability: Service quality can vary based on who provides it, when, and where.
 - Perishability: Services cannot be stored; they are time-sensitive (e.g., a missed flight).

WHAT IS SERVICE DESIGN?

- Service design focuses on improving services to **better meet the needs** of users and customers. It examines all aspects, including activities, infrastructure, communication, and personnel, to enhance service quality and interactions.
- The goal of service design is to develop front and back office strategies that effectively address customer needs while being **economically sustainable** for providers. Ideally, services should be user-friendly and competitive in the market.

“When you have two coffee shops right next to each other, and each sells the exact same coffee at the exact same price, service design is what makes you walk into one and not the other.”

SERVICE DESIGN EXAMPLE -- STARBUCKS

What value did Starbucks provide to customers? Why users come to Starbucks?

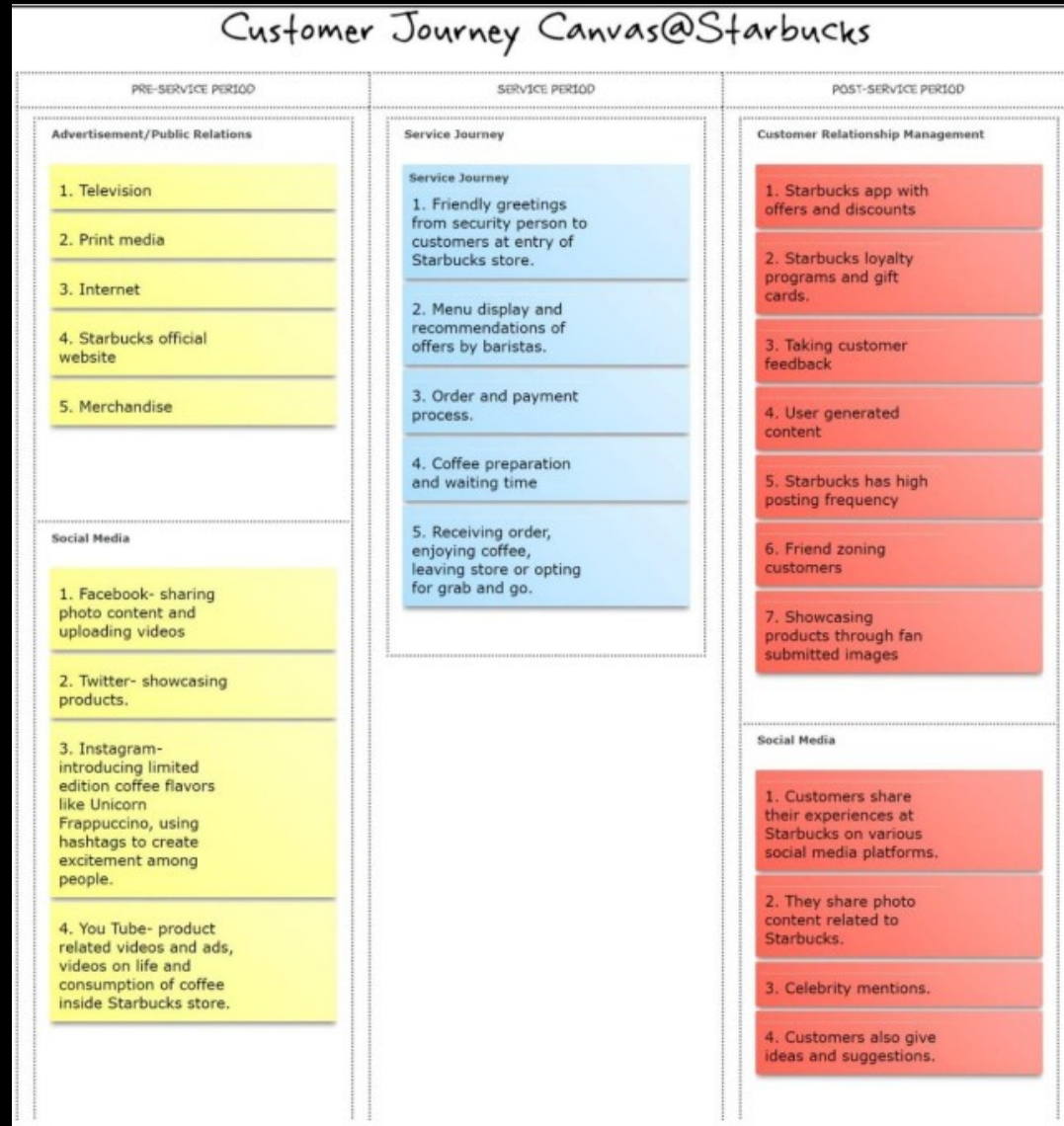
Functional: Quality, Variety, Sensory Appeal

Emotional: Reduce Anxiety, Design Aesthetics, Badge value

Relational: Provides Hope, Affiliation, Belonging



SERVICE DESIGN EXAMPLE -- STARBUCKS



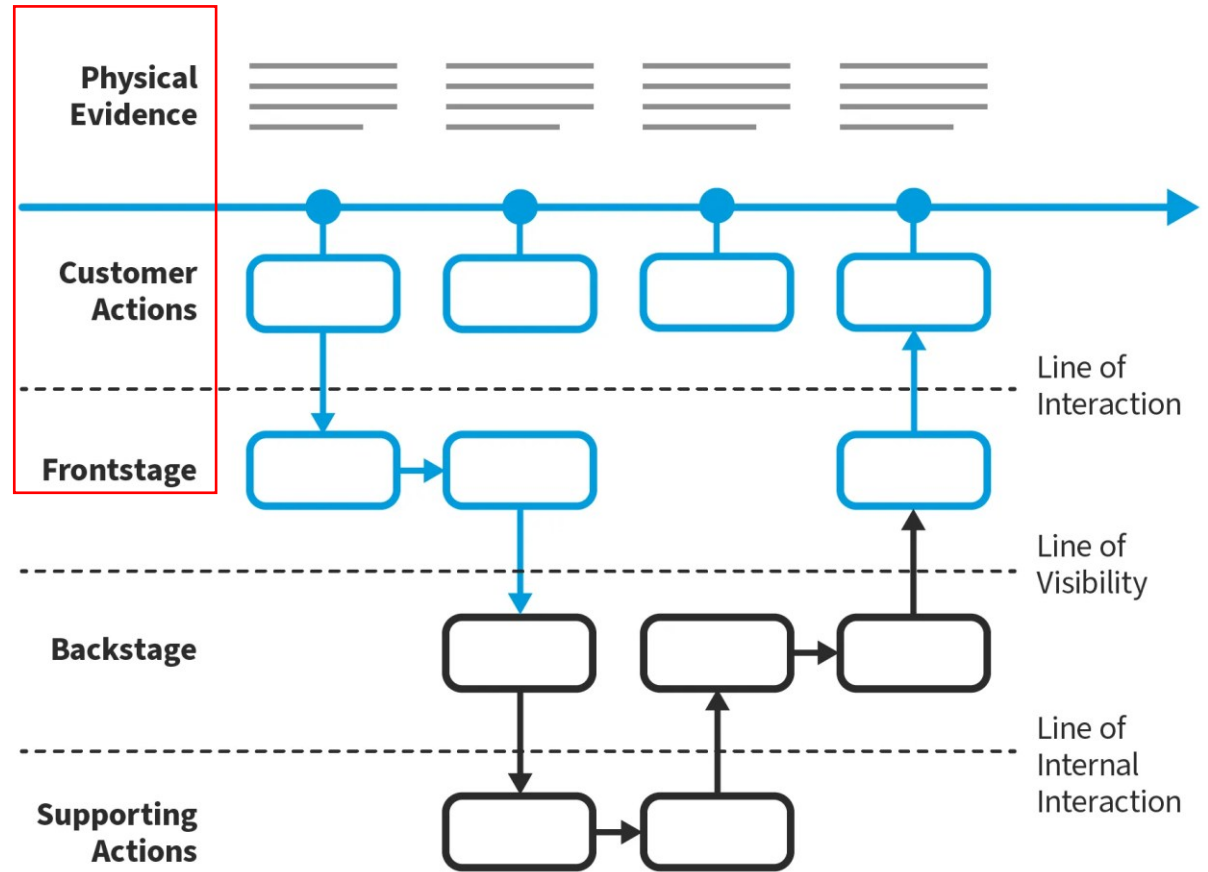
A **user journey** is a visual or narrative representation of the steps a user takes to accomplish a specific goal when interacting with a product or service. It outlines the user's experience from start to finish, capturing their emotions, thoughts, and actions at each stage.

SERVICE BLUEPRINT

- A **service blueprint** expands on a customer journey map. It details every interaction a customer has with an organization during their lifecycle.
- This blueprint goes deeper and examines all supporting physical and digital interactions. It often appears as a diagram with swim lanes. Each lane represents a specific category. Arrows link interactions between lanes to show the workflow.

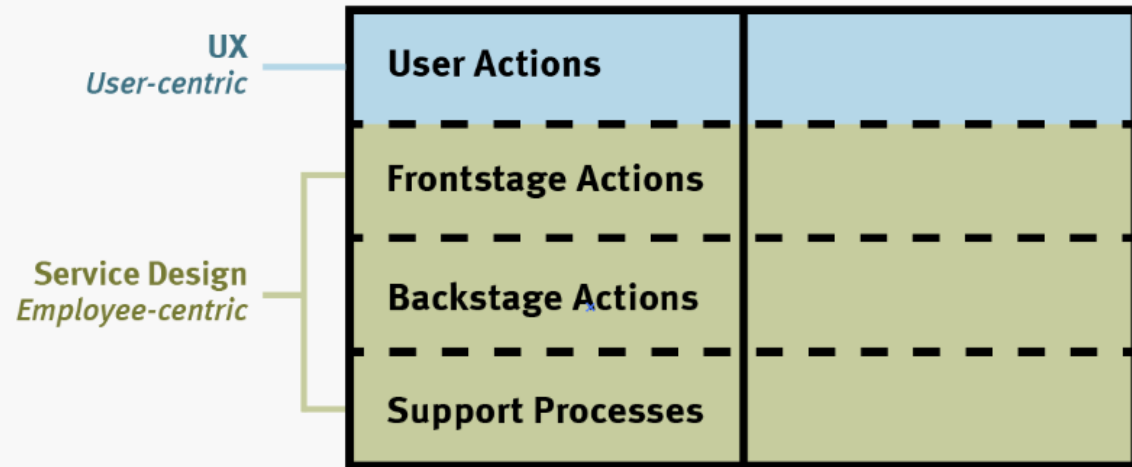
Service Blueprint Anatomy

Touchpoints



UX VS. SERVICE

Structure of a Service Blueprint



NNGROUP.COM NN/g

User Experience vs. Service Design



NNGROUP.COM NN/g

SERVICE DESIGN FOR HKUST RECYCLING

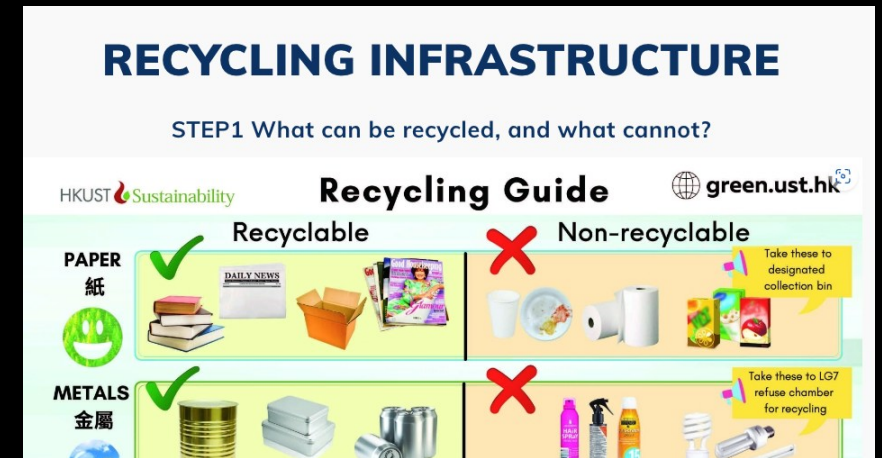
Some physical evidences:



User Journey



Service blueprint



Recycling Infrastructure | HKUST Sustainability

Further improvement

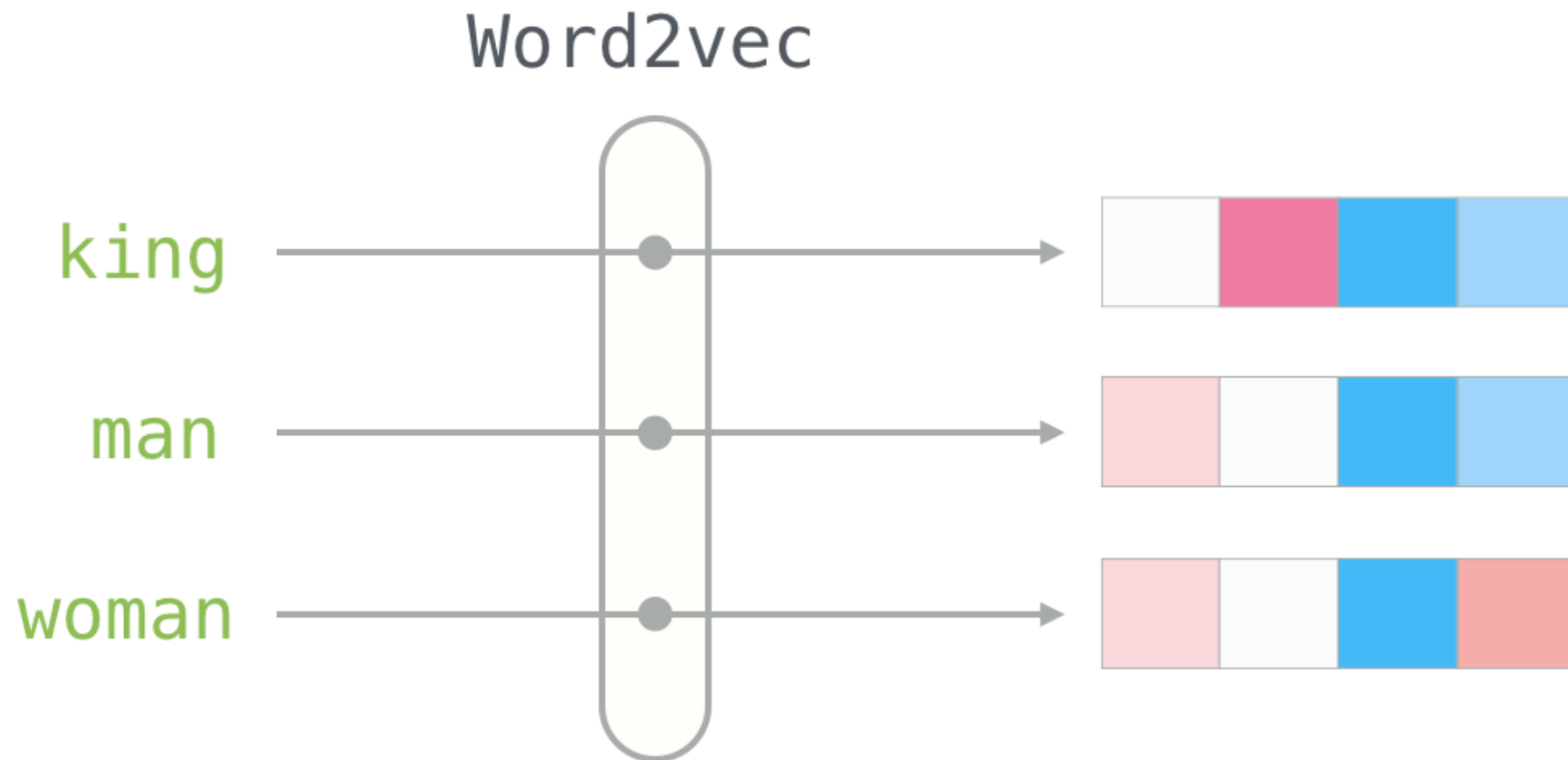
WHAT IS TRANSFORMER

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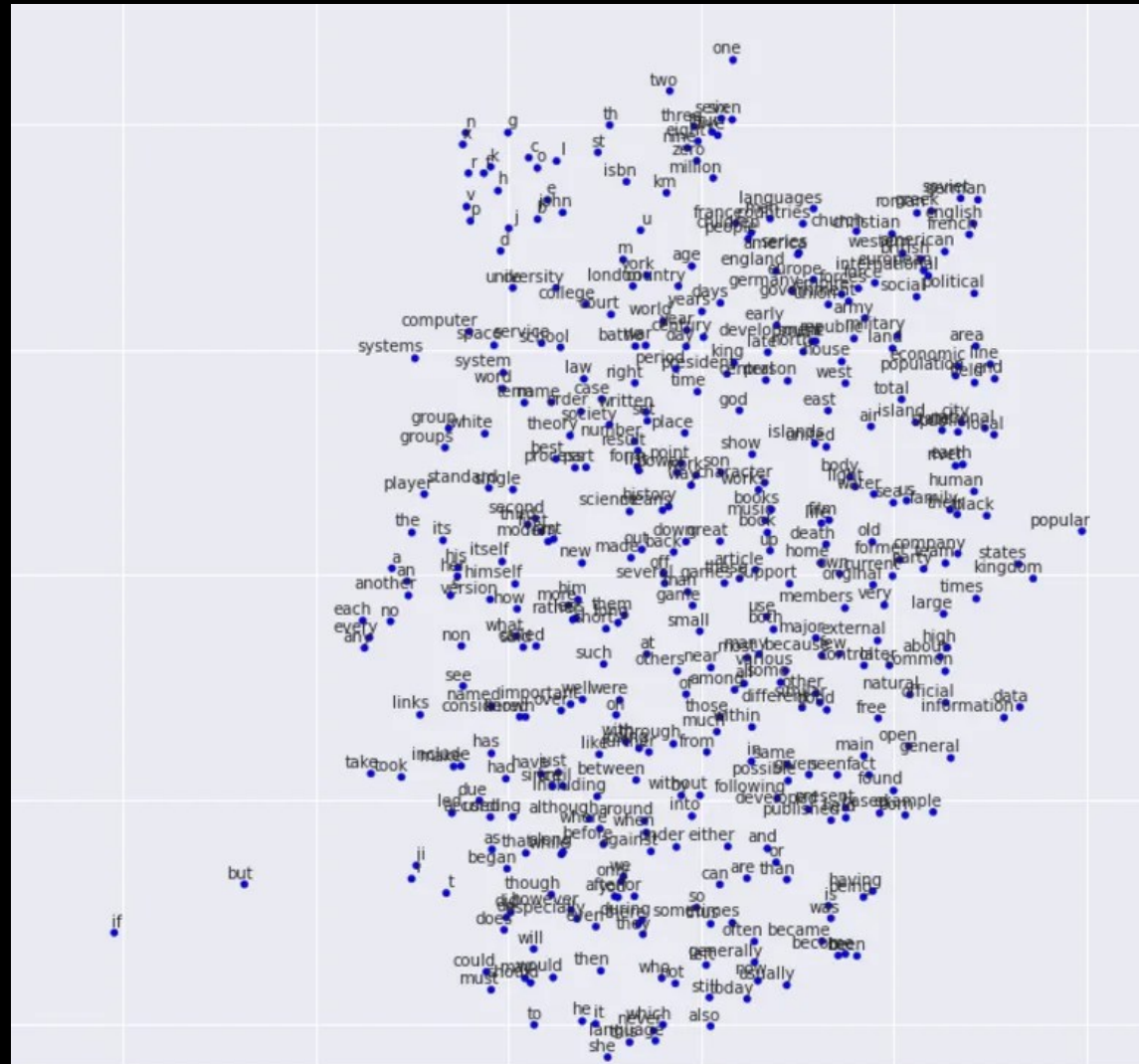
CAN WE DO THIS MORE EFFICIENT?

- Word2Vec



CAN WE DO THIS MORE EFFICIENT?

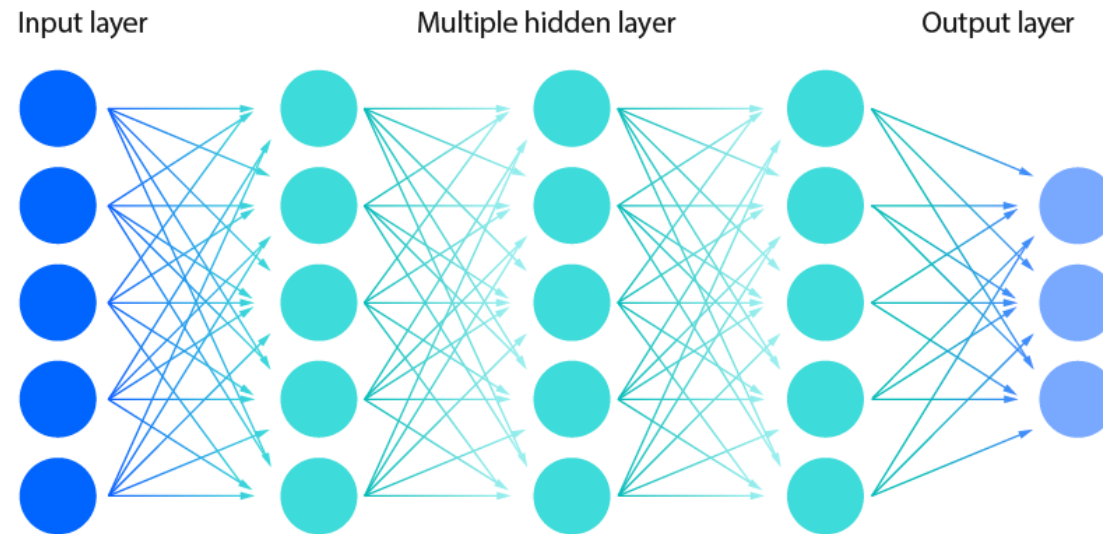
- Word2Vec



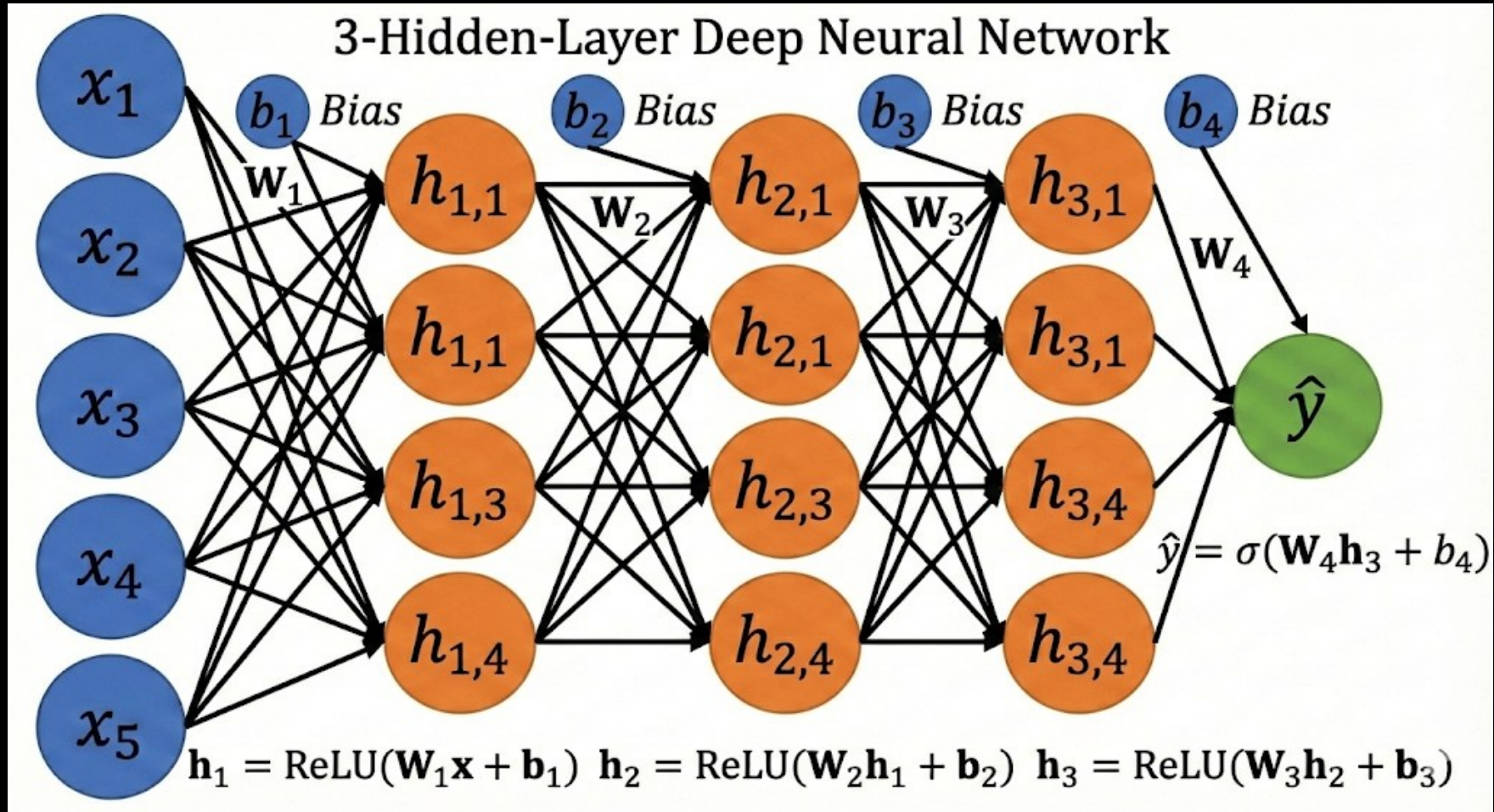
HOW TO INPUT TEXT TO NEURAL NETWORKS

- Tokenizer
- Word2Vec

Deep neural network



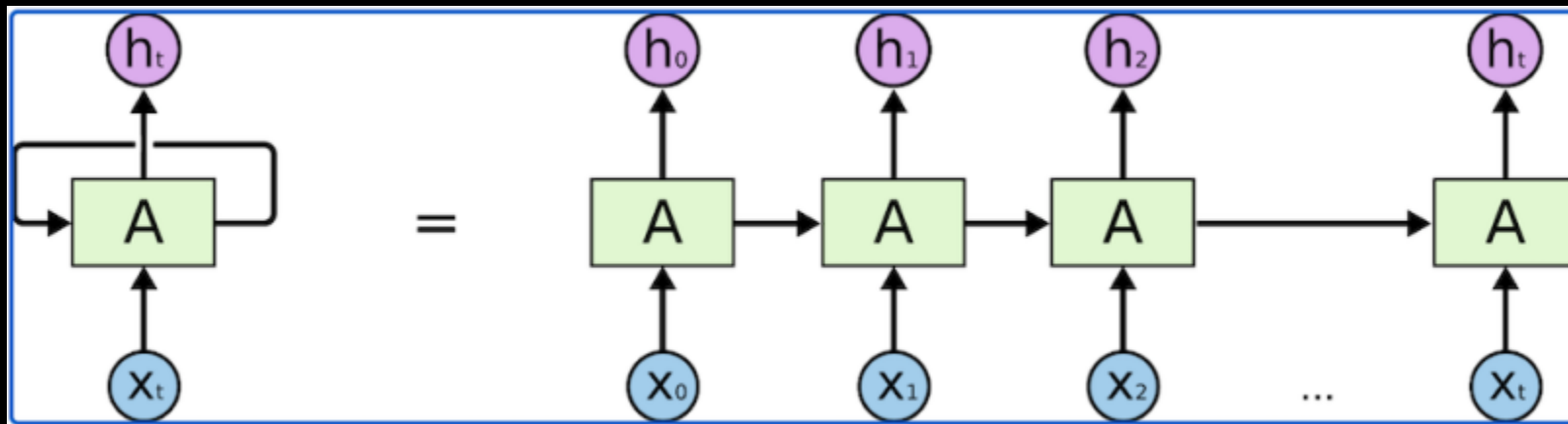
WHAT WE HAVE DEFINED



This cannot process unfixed length sequences!

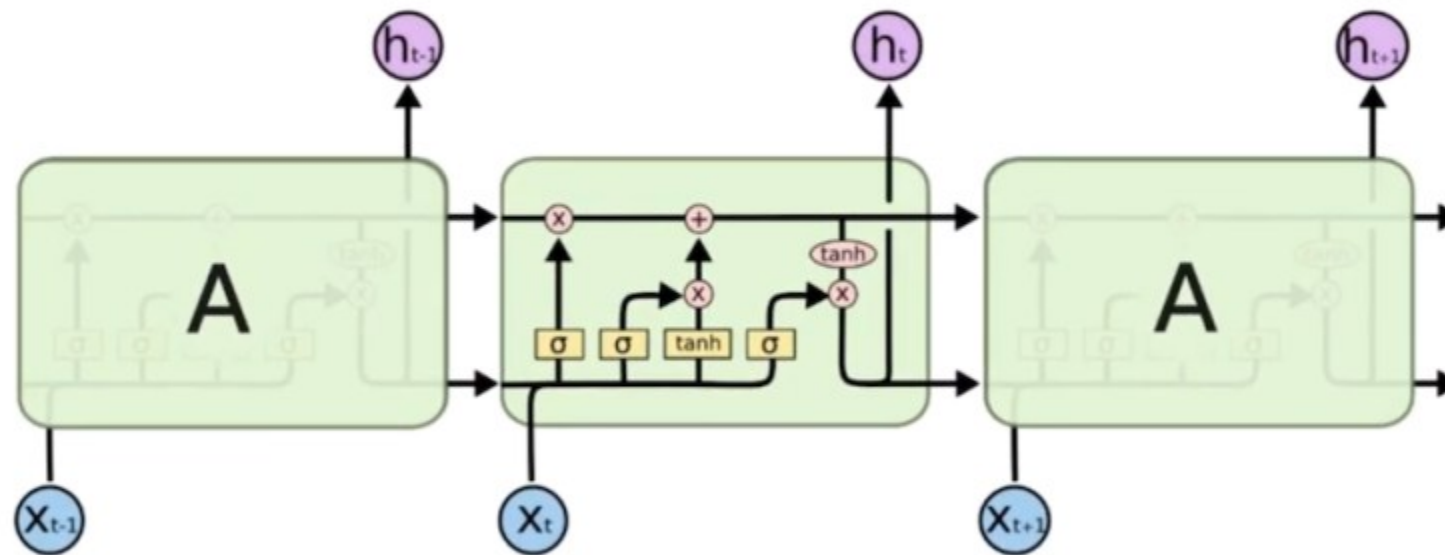
HOW TO PROCESS SEQUENCES?

- Recurrent Neural Network



HOW TO PROCESS SEQUENCES?

- Long short-term memory (LSTM)



The repeating module in an LSTM contains four interacting layers.

HOW TO PROCESS SEQUENCES?

- Attention is all you need!

Attention is all you need

[A Vaswani, N Shazeer, N Parmar...](#) - Advances in neural ..., 2017 - [proceedings.neurips.cc](#)

... to attend to **all** positions in the decoder up to and including that position. **We need** to prevent ... **We** implement this inside of scaled dot-product **attention** by masking out (setting to $-\infty$) ...

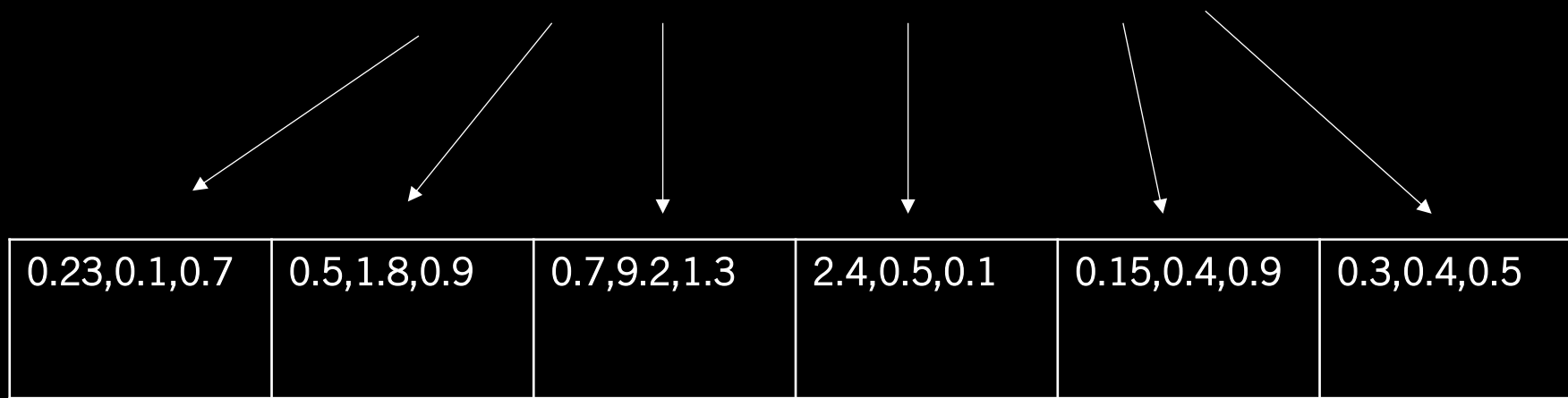
☆ 保存 引用 被引用次数: 235261 相关文章 所有 71 个版本 》》

HOW DO WE GET AN ANSWER?



TOKENIZATION

What is the weather like?



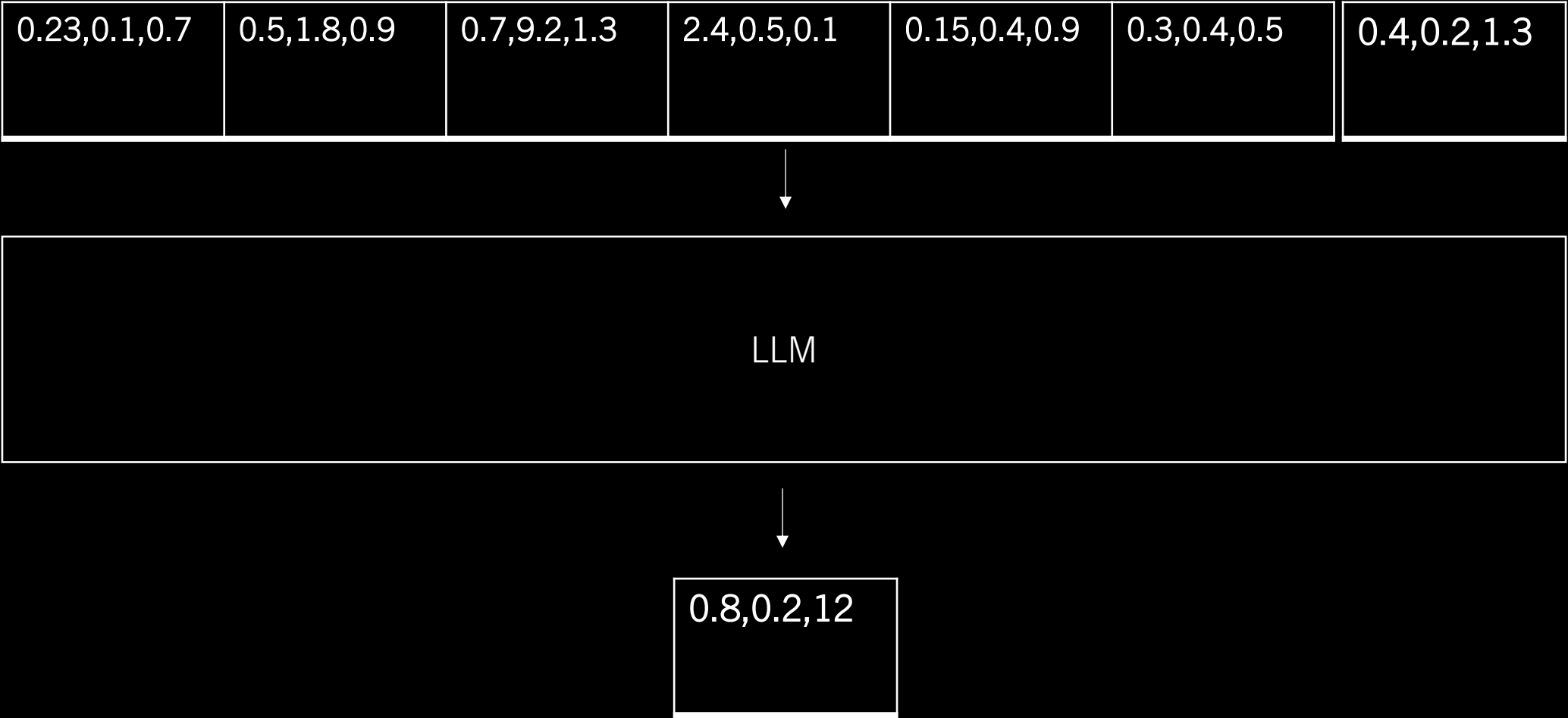
DECODING PHASE

0.23,0.1,0.7	0.5,1.8,0.9	0.7,9.2,1.3	2.4,0.5,0.1	0.15,0.4,0.9	0.3,0.4,0.5
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0.4,0.2,1.3

DECODING PHASE



DECODING PHASE

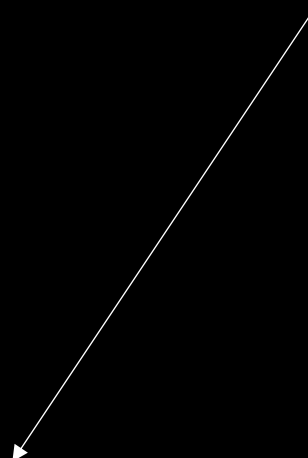
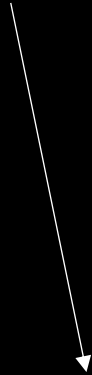
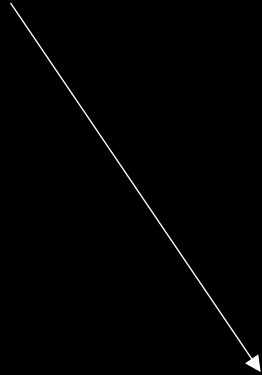
...	0.5,1.8,0.9	0.4,0.2,1.3	0.8,0.2,12	0.2,0.2,0.8	1.5,0.9,2.1
-----	-------------	-------------	------------	-------------	-------------



-1,-1,-1

<EOS> End of Sequence token

0.4,0.2,1.3	0.8,0.2,12	0.2,0.2,0.8	1.5,0.9,2.1
-------------	------------	-------------	-------------



It's rainy outside.



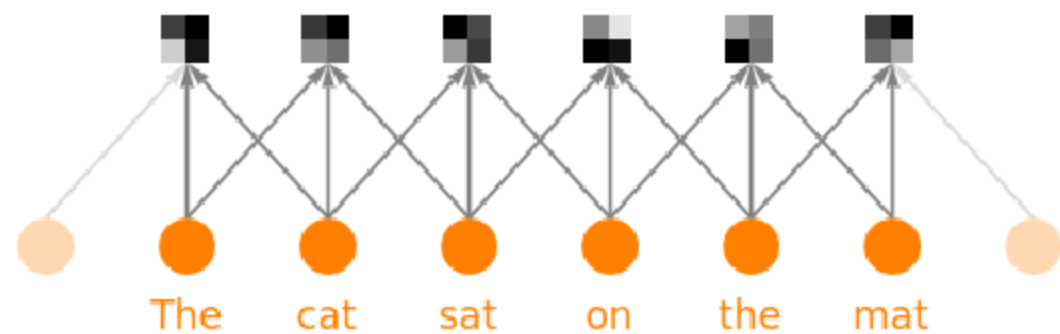
recite	the	first	law	\$									
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the

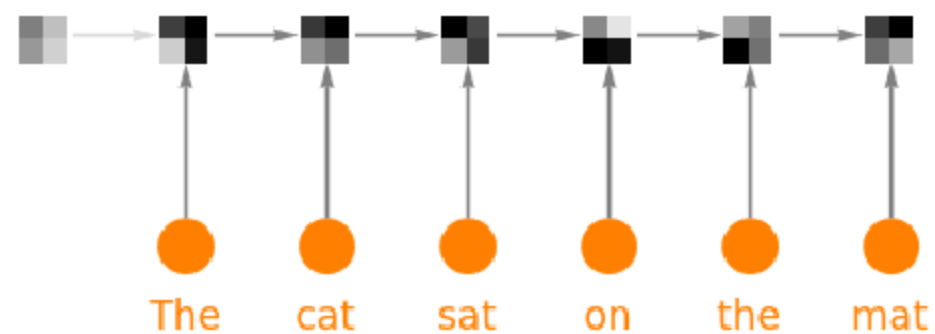
law

\$

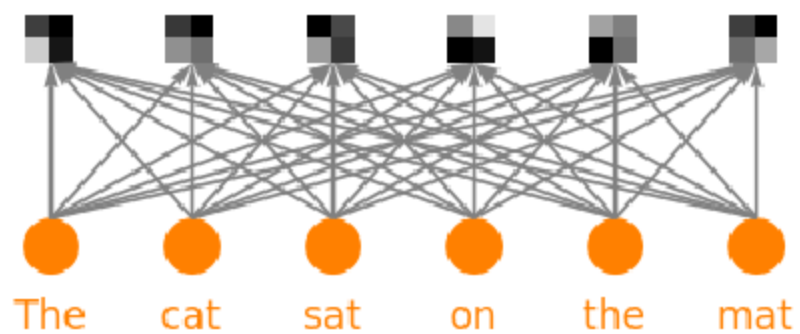
Convolution



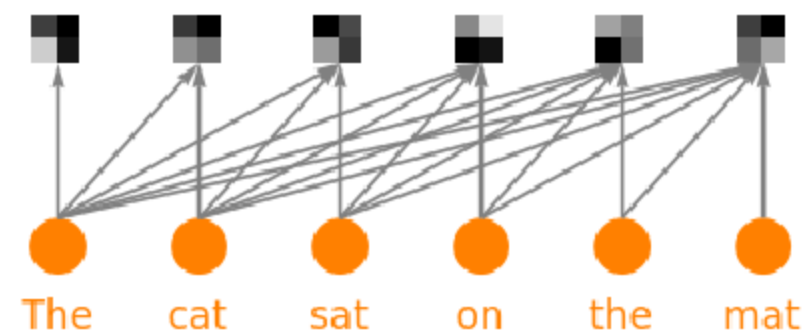
Recurrence



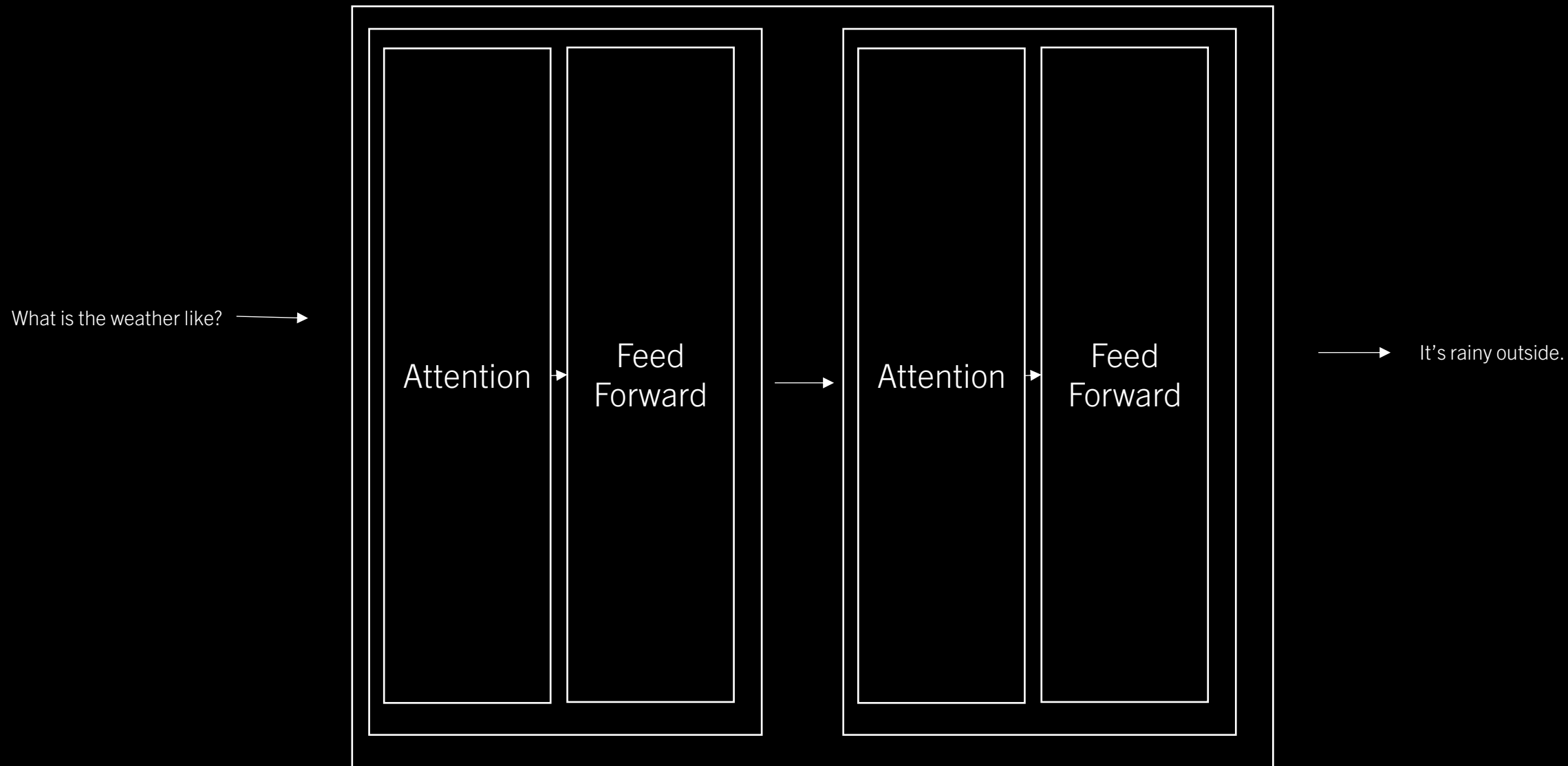
Self-Attention



Causal Attention



LOOKING A BIT DEEPER



ATTENTION

- LLM predicts the next word given the previous sentences.
- However, not all words contribute equally to the prediction of the next word.
- We need a way to mark an importance of words in a sentence.

The cat likes to drink _____ milk _____.

ATTENTION

- In an attention layer, each word gets three tensors: Query, Key, and Value

The

cat

likes

to

drink

milk

.

The

cat

likes

to

drink

milk

.

ATTENTION

Query Key Value



- For each word, we multiply the query with all other keys to get the attention weights.
- This represents the importance of the other words.

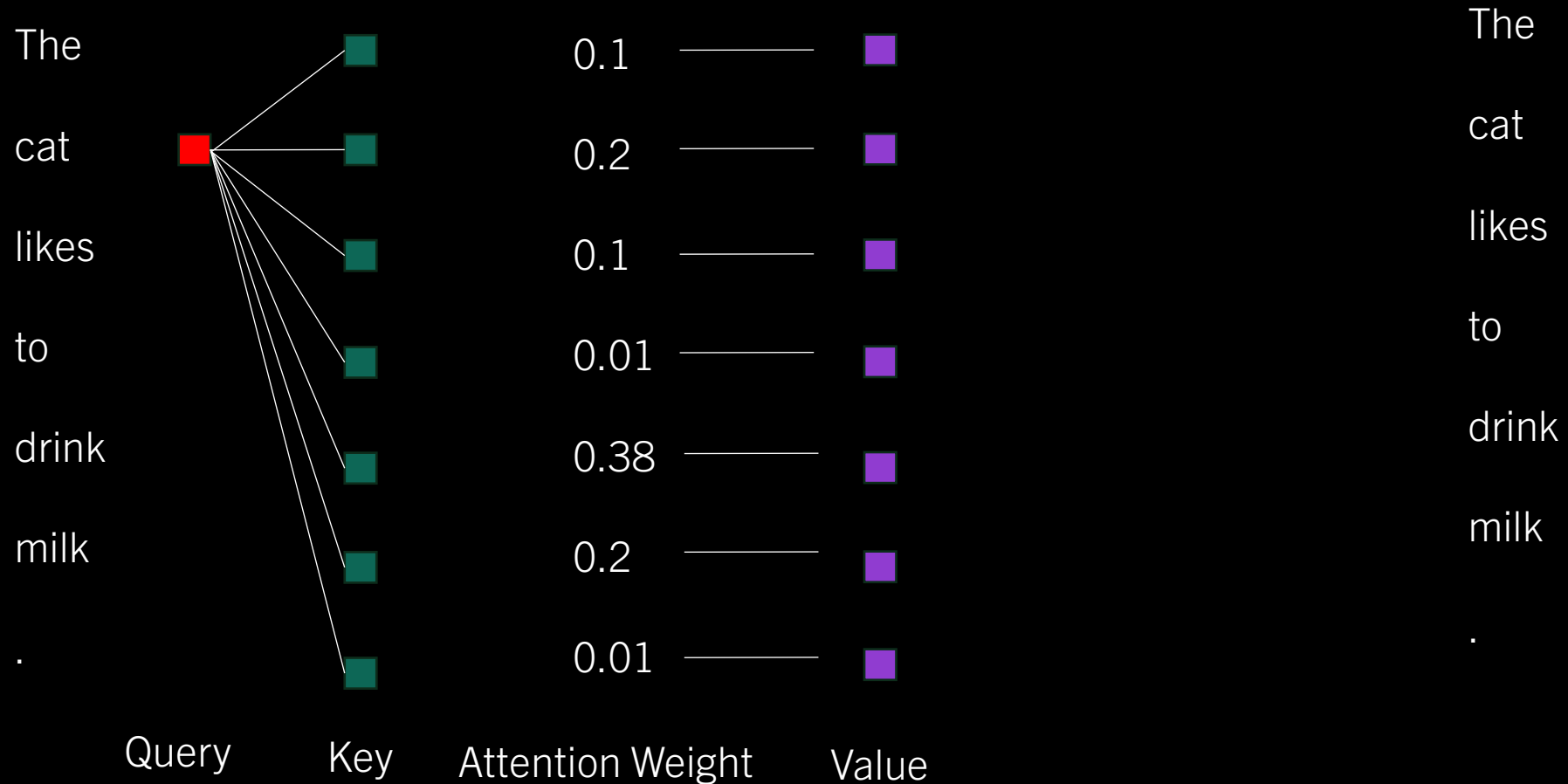


ATTENTION

Query Key Value



- Then we multiply each value with there corresponding weights and add them together.
- That will be the new tensor that will be passed on for that word.

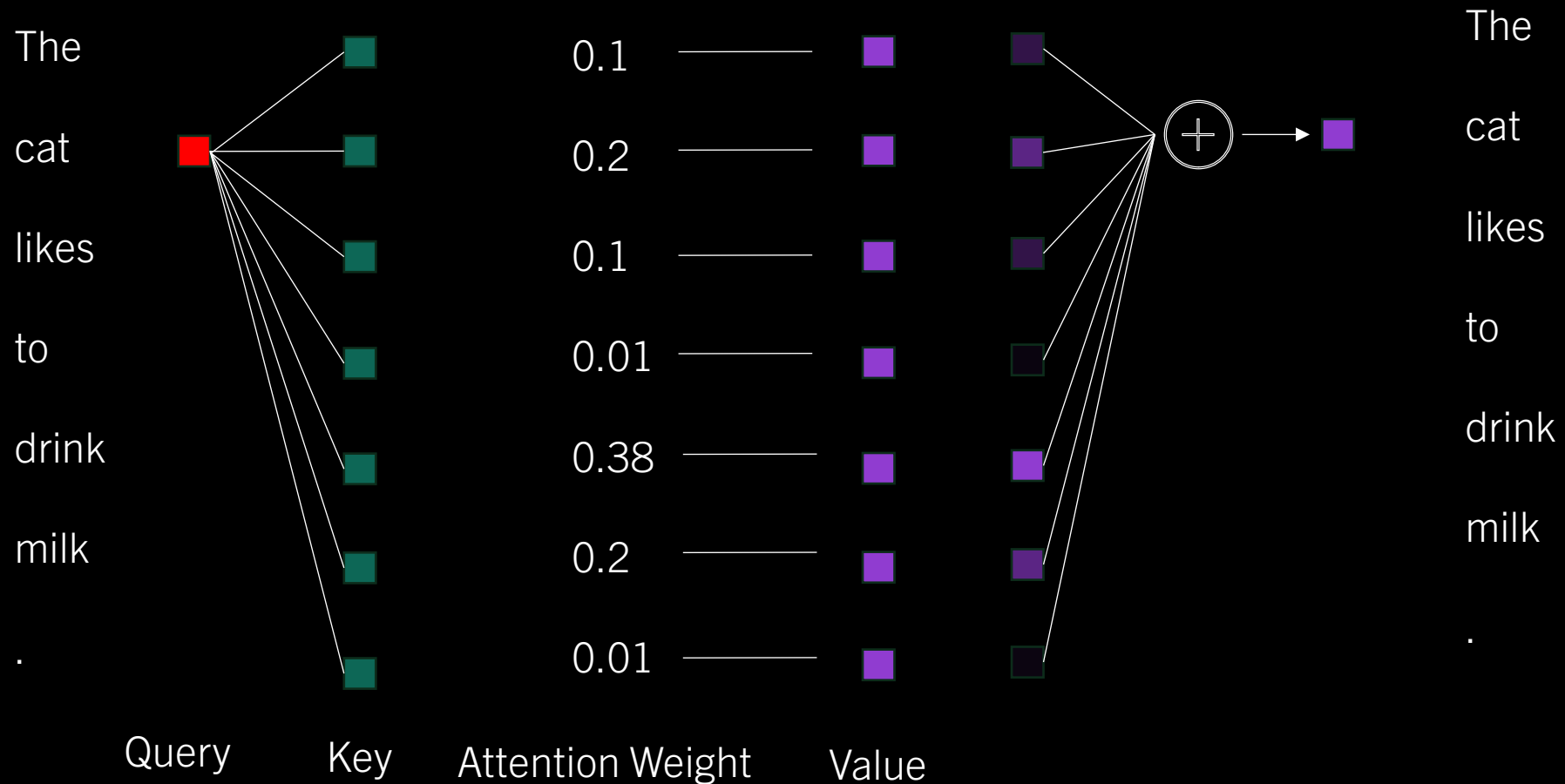


ATTENTION

Query Key Value



- Then after each layer, the latent value of each word will be updated in relation to the importance of surrounding words.



ATTENTION



Query



Key



Value

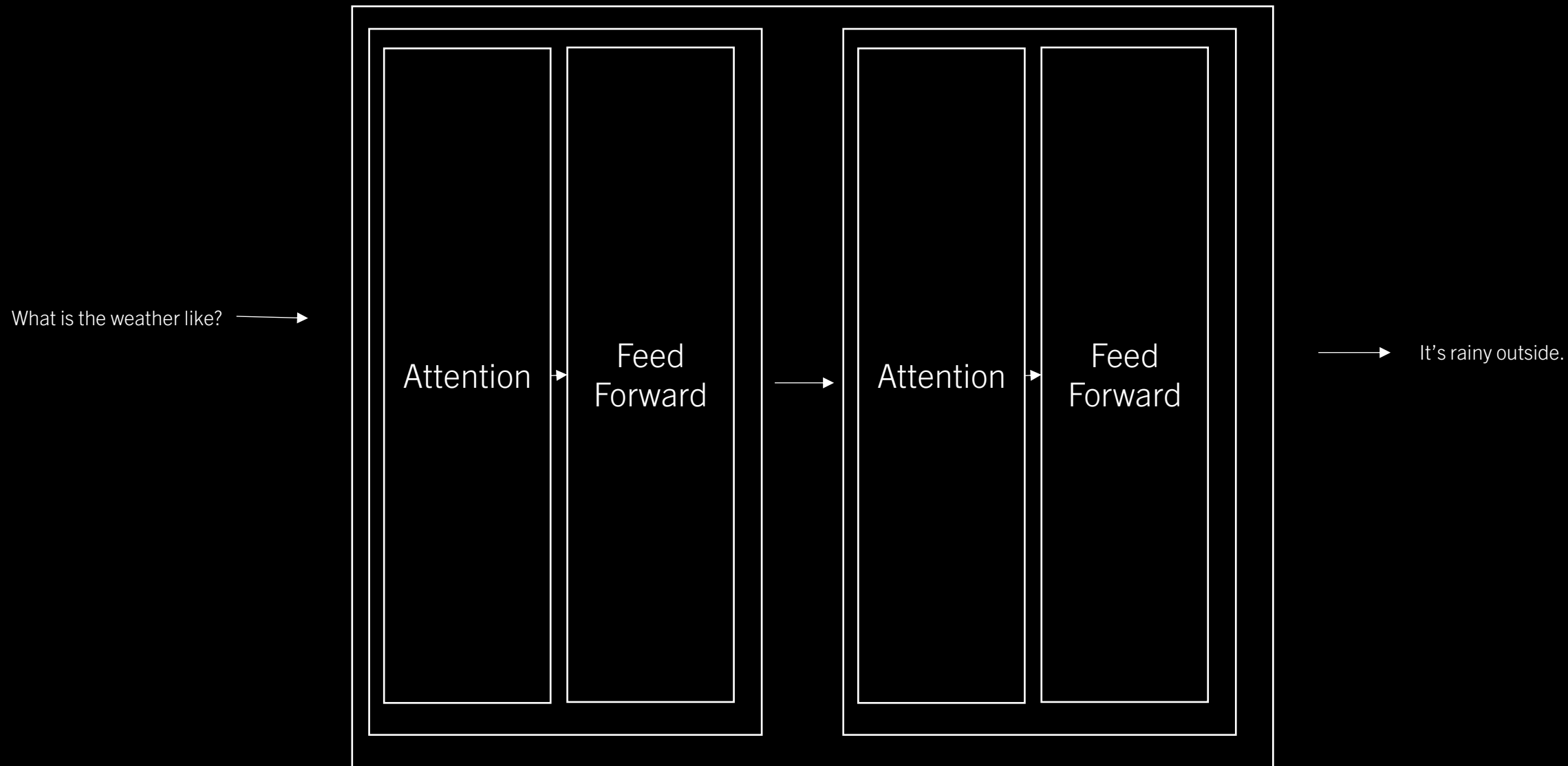


Updated Output

WHY IS ATTENTION SO POWERFUL?

- It allows the neural network to learn the **importance of the relationship** of the dataset.
 - The attention mapping spans the entire data sequence, so there are no blind spots.
 - Like filters for CNN, we can have **multiple attention heads**, to tackle different information.
 - This architecture seems to **train easily** even when the layers get deeper and wider.
-
- Notice that in the example the Q,K,V were all from the same source. We can actually have the Q, K, V come from different sources, and merge outputs from different layers!
 - This is called **cross-attention**, and this is how we can inject text prompts to diffusion models!

FEED FORWARD LAYER



TODAY

- What is UI design
 - What is UX design
 - What is service design
-
- Technical part – what is a Transformer

MINI PROJECT – MORE DETAILS RELEASED ON FRIDAY LAB

- Designing a Personality-Driven Virtual Character with Visuals
- Distinct Personality (LLM): Design a chatbot with a highly recognizable and consistent personality.
- Visual Identity & States (Midjourney): Use Midjourney to generate the character's visual design.
- Image Triggering Mechanism: Design an interaction mechanism where the chatbot can "reply" with images.

MINI PROJECT – MORE DETAILS RELEASED ON FRIDAY LAB

- Formulate a group, also for your final project.
- 11 students – 5 groups – one group could have 3 students.
- <https://forms.gle/egpUoNTSpXSSomrJ9> DDL: March 13, before our Friday lab.
- Balance the background!



	Session 1 (4:30–6:20 pm, Monday)	Session 2 (3:00–4:50 pm, Friday)	
W1	Introduction & Overview	History of AI & Design	
W2	From Deep Learning to Generative AI	Environment Setup & Python Primary & APIs	A1: text-based image / Generation Test
W3	Chinese New Year Eve	Online Office Hours	Chinese New Year
W4	Design Thinking with AI	Image Generation Lab	A2: Card Design
W5	Brand design with AI (+Customizing Image Gen and Image Control)	Brand Design Lab	Group formulation A3: Brand Design
W6	UI & UX & Service Design (+Basics for language processing, architecture of Gen AI) Card Design presentation	UI & UX Design Lab (UI Design)	Mini project release A4: UI Design
W7	Prototype with AI (+Vibe coding) Brand Design presentation	Prototype AI I LLM Prompt)	
W8	Prototype AI II (LLM Processing + CoT + Tools Calling)	RAG & Vector DB UI & UX presentation	Final project release
W9	Agent AI I (RAG & Vector DB Lab)	Mini Project Presentation	Mid-term break
W10	Agent and Human-AI Cooperation	Agent AI II Lab (Multi Agent)	A5: Agent Design
W11	Final Project Proposal	AI Ethics Agent Design presentation	
W12	AI Safety Lab	Labor Day holiday	
W13	Final Project Presentation	Final Project Presentation	

THANK YOU!



AI for Design

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